

Machine Learning for Integrating Risk Factors in Osteoporosis Prediction

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OSTEOPOROSIS RISK PREDICTION USING MACHINE LEARNING AND BIOMARKERS

Abstract

Osteoporosis, a systemic skeletal disorder, is marked by low bone mineral density (BMD), disrupted bone turnover, and heightened fracture risk. The gold standard for osteoporosis diagnosis, dual-energy X-ray absorptiometry (DXA), quantifies areal BMD (aBMD) but fails to account for bone microarchitecture and the associated biological mechanisms. Traditional clinical risk prediction models and screening tools, including FRAX, Garvan Fracture, and QFracture, have moderate predictive performance, commonly underestimate non-hip fractures, and lack extensive external validation. This review examines the application of machine learning (ML) methods with clinical risk factors and biomarkers to enhance osteoporosis-related fracture risk prediction. Genetic and biological determinants, DXA and three-dimensional (3D) QCT imaging, supervised and unsupervised ML models, and deep learning neural networks, as well as multimodal, cross-sectional, and longitudinal data integration, are discussed. Supervised ML models yield area under the curve (AUC) values up to 0.835, while deep learning models yield femoral neck prediction AUCs of 0.851 and 0.827. Multimodal hybrid methods combining HR-pQCT imaging with clinical variables improve prediction to approximately 0.90 AUC; in comparison, DXA assessment scores are around ~0.71. Limitations include overfitting, population bias, and low interpretability. ML methods can significantly improve osteoporosis risk predictive accuracy, particularly when combined with multiple clinical and biomarker variables; however, traceability and data limitations pose barriers to clinical translation. Prospective research and development should emphasize multimodal, biomarker-based longitudinal datasets, hybrid interpretable methods with feature attribution, and comprehensive external validation to accelerate clinical integration.

BACKGROUND

Demographics & Public Health Burden

Over 200 million people worldwide suffer from osteoporosis, 70% of whom are women.¹ According to the International Osteoporosis Foundation, one in three women and one in five men over the age of 50 are affected by the disease.² An even greater number of individuals suffer from low bone density,³ which increases the risk of progression to osteoporosis. The largest at-risk demographic group is postmenopausal women, largely due to reduced estrogen levels after menopause. Women also tend to experience bone loss earlier in life than men and lose bone at a faster rate, contributing to a biologically driven gender disparity in osteoporosis risk.⁴ Between 2007 and 2018, the percentage of women diagnosed with osteoporosis increased from 14% to over 19%, while no significant change was observed in men.³ However, osteoporosis in men has a higher proportion of undiagnosed cases and is associated with a higher mortality rate, potentially due to a later age of first fracture.⁵ In men, a large proportion of osteoporosis cases are secondary in development, with primary causes including but not limited to hypogonadism (associated with reduced estrogen levels),⁴ alcoholism, and glucocorticoid overuse.⁵ Secondary osteoporosis is comparatively less common in women. For a given bone mineral density measurement, however, male and female fracture risk were proven to be almost equal.⁵ As such, both men and women remain vulnerable once bone loss is present.

The public health burden of osteoporosis is expected to increase significantly within the following decade. By 2030-2034, approximately 263 million individuals will be affected by the disease.¹ This trend is in line with an aging global population and increased life expectancies. According to the World Health Organization, 2.1 billion individuals, or 22% of the world's population, will be 60 or older by 2050; in comparison, there were 1 billion people in this age bracket in 2020.² An aging population will increase the importance of geriatric care and shift healthcare focus toward early detection and prevention of age-related chronic diseases. Currently, the healthcare market for osteoporosis is estimated at around \$100 billion worldwide, and over \$30 billion in the United States alone;³ these numbers are expected to increase. Closely following the broader healthcare market, the global drug market for osteoporosis is projected by industry reports to be worth over 20 billion by 2032.⁴ Additionally,

annual fracture rates in the United States are projected to climb from 1.9 to 3.2 million (68%) from 2018 to 2040, and fracture-related costs are expected to increase from \$57 to \$95 billion.⁵

Figure 1 | Prospective growth of the osteoporosis drug market.

Note: Omitted from this public version due to copyright. Please refer to Surwase⁶ for the original figure.

The Silent Disease

Osteoporosis is a bone disease characterized by low bone mass and reduced bone mineral density (BMD). Often termed “the silent disease”, osteoporosis usually remains undiagnosed until the patient experiences a fracture. Research shows that approximately 69.12% of affected individuals remain undiagnosed, and that number increases to 84.77% for individuals ages 50 through 59.⁵ Factors contributing to these numbers include asymptomatic progression (including silent fractures), untreated minor fractures, and overlap with normal age-related bone loss. A simple vertebral fracture can easily go unnoticed by the patient, yet increases the likelihood of subsequent hip fractures two- or threefold.⁷ In turn, a hip fracture increases the likelihood of further fractures, including wrist fractures, by 2.5-fold.⁷ The silence of this chain reaction is a key reason why an osteoporosis diagnosis can seem unexpected to the patient, while in reality, the disease was progressing for a long time - possibly even decades. Age-related bone degradation can begin as early as age 40; according to researchers at the University of Sydney, bone resorption (metabolic breakdown) starts occurring faster than bone formation in the fourth decade of life.⁸ Although rarer in younger patients, osteoporosis can occur at any age in individuals with genetic risk factors; others may develop idiopathic osteoporosis without a clearly identifiable cause.

While osteoporosis progresses unnoticed, an alarming number of patients fail to recover functional mobility after experiencing their first fracture. Following a first major fracture, susceptibility becomes secondary to clinical outcome, with both men and women at risk of irreversible functional decline. Vertebral fractures are related to chronic pain and spine deformity, while 60% of hip fracture patients never fully regain prior independence.⁷ In addition, self-reported adverse patient consequences include reduced psychological well-being and perceptions that physicians did not adequately inform them about maintaining bone health. One patient reported that they “became ill with rheumatoid arthritis in my early teens... [healthcare providers] started talking about osteoporosis only 20 years later”,⁹ suggesting that their condition was overlooked until they reached a certain age group. Late recognition and the absence of early warning signs, combined with the asymptomatic nature of the condition, contribute to its “silent” progression.

NEED

Risk Prediction

The high incidence of osteoporosis, its growing healthcare burden, and its asymptomatic progression together place increasing pressure on early identification and preventive care. Osteoporosis diagnostic and risk prediction tools aim to meet this clinical need; however, these tools do not consistently identify all patients at risk. The clinical standard for osteoporosis diagnosis is a bone mineral density (BMD) T-score equal to or below -2.5 standard deviations, yet the majority of individuals who experience their first fragility fracture do not meet this threshold.¹⁰ A 2025 study evaluated the performance of three simple osteoporosis risk assessment tools – the Osteoporosis Risk Assessment Instrument (ORAI), Osteoporosis Risk Assessment Tool (OST), and Osteoporosis Index of Risk (OSIRIS) – to predict disease in healthy, younger postmenopausal women aged 50-64.¹¹ The researchers found “less than ideal” performance in all three risk assessment tools; the most accurate of the three, ORAI, yielded a moderate accuracy of 66.3%.¹² These findings suggest that factors beyond the common clinical risk indicators – including age, estrogen use, fracture history, and low body mass index (BMI) – contribute to fracture risk and highlight the need for further evaluation of osteoporosis risk prediction tools.

Preventive care plays a vital role in reducing the incidence of osteoporosis when implemented early. It often includes weight-bearing and resistance exercise, monitoring and maintenance of healthy vitamin D, estrogen, and calcium levels, lifestyle modifications such as fall-prevention strategies in the home, visual impairment correction when necessary, cessation of tobacco use and/or alcohol overuse, and adjustment of psychotropic medication when appropriate.⁷ Progressive resistance training (PRT) is often recommended for at-risk postmenopausal women; it may increase femoral neck BMD.¹³

Pharmacologic therapy is primarily used to reduce the risk of future falls and fractures in patients who have previously experienced a fragility fracture. However, it is also implemented as preventive therapy for high-risk individuals at the discretion of a healthcare provider. Anabolic medications rebuild lost bone mass by activating PTH receptors. Bisphosphonates, which inhibit osteoclast-mediated bone resorption, slow bone loss and help preserve bone microarchitecture, thereby increasing BMD. Several bisphosphonates have been approved for preventive therapy; these include alendronate sodium, ibandronate sodium (oral preparations), risedronate sodium, and zoledronic acid. Alendronate has been proven to reduce the incidence of vertebral fractures by 48 % over a 3-year period for patients without a previous vertebral fracture, while ibandronate reduces vertebral fractures by 50% over the same time period.⁷ Preventive dosing is typically lower than treatment dosing. Timely initiation of preventive interventions is crucial for at-risk patients, especially younger postmenopausal women, who may experience bone loss of up to 4% per year.¹⁴

Pathophysiology & Risk Factors

Osteoporosis is a complex, multifactorial disease that is driven by a web of underlying biological mechanisms. The physiological changes associated with osteoporosis are usually asymptomatic, contributing to the lack of clinical signs and, frequently, underdiagnosis. These changes are characterized by a disruption of bone homeostasis and gradual bone matrix degradation. Osteoclasts, the cells responsible for bone resorption, are typically overactive in osteoporosis and break down the bone matrix faster than osteoblasts can replace it. Estrogen deficiency, low calcitonin, and increased PTH (parathyroid hormone) levels are all directly associated with osteoclast overactivity.¹⁵ Besides hyperparathyroidism, low vitamin D and high phosphate levels also contribute to high PTH levels, indirectly promoting osteoclast overactivity.

Figure 2 | Multisystem physiological contributors to osteoporosis risk.

Note: Omitted from this public version due to copyright. Please refer to Wang et. al.¹⁶ for the original figure.

In addition to hormonal regulation, cellular aging also contributes to the disruption of bone homeostasis. Osteocytes, which make up 90-95% of total bone cells,¹⁷ play a crucial role in the regulation of bone mineral homeostasis. As bone tissue ages, osteocytes are particularly sensitive to senescence - i.e., a state where cells stop dividing, but do not die. Senescent osteocytes release SASP factors. The senescence-associated secretory phenotype (SASP), a pro-inflammatory secretory state, leads to the release of inflammatory cytokines and dysregulated signaling within the bone microenvironment, which leads to slower bone remodeling and net bone loss.¹⁸ SASP-associated inflammatory mediators are increasingly considered as potential biomarkers of altered bone remodeling and osteoporosis progression.

An estimated 60-80% of bone mineral density variability is inherited, while 20-40% is the result of environmental factors.¹⁹ However, the genomics behind the heritability of the disease is not fully understood. Researchers have identified SOST and LRP5 as two genes encoding proteins that regulate the Wnt signaling pathway, crucial for bone homeostasis. LRP5 protein deficiency has been found to lead to osteoporosis-pseudoglioma, while SOST protein deficiency results in bone gain.²⁰ Structural integrity of the bone matrix is affected by gene COL1A1, which encodes type I collagen, the primary organic component of bone matrix. A landmark study from the University of Aberdeen found that polymorphism in the COL1A1 Sp1 binding site increases COL1A1 expression and alters collagen protein composition, resulting in reduced bone strength and higher fracture risk. Finally, genes involved in hormonal and cytokine signaling include ESR1 (estrogen receptor 1 gene) and TNFRSF11B, the gene which encodes osteoprotegerin (OPG), a receptor protein that inhibits bone resorption. The absence of ER α , the protein encoded by ESR1, has been found to affect the periosteum and lead to reduced cancellous bone mass.²¹ The

ESR1 and TNFRSF11B genes co-regulate osteoclast activity and are crucial for bone remodeling and overall bone health.

CURRENT APPROACHES

DXA-Based Diagnosis

Dual-energy X-ray absorptiometry (DXA), introduced for clinical use in the 1980s, is the gold standard for measuring BMD and clinically diagnosing osteoporosis. DXA scans generate 2-dimensional scans of bone structures and report BMD (usually in units of grams/cm^2) by dividing bone mineral content (BMC) by the bone area region of interest (ROI). X-ray beams attenuate bone tissue and reflect the density thereof. The method involves measuring bone density, usually of the hip (femoral neck) or spine (lumbar area), but not bone size or strength. T-scores, which benchmark an individual's bone mineral density relative to that of a healthy young adult, are used for purposes of diagnostic classification for postmenopausal women or patients over the age of 50. These values are calculated using normative reference databases stratified by demographic factors such as sex, age, and race.¹⁰ The World Health Organization considers a T-score ≤ -2.5 at the lumbar spine, total hip, femoral neck, or one-third radius consistent with osteoporosis,²² while a T-score ≤ -1.0 is considered consistent with osteopenia. For patients under the age of 50, Z-scores are generally preferred. Z-scores benchmark bone mineral density relative to that of healthy persons of the same age, sex, and ethnicity,²³ which enables identification of low BMD relative to the demographic group. A Z-score ≤ -2.0 is defined as lower than average BMD in younger men and premenopausal women.²² Pediatric values are calculated using a pediatric normative base.¹⁰

In addition to diagnosis, DXA scans are used to monitor bone health and treatment effects in osteoporosis patients. Significant changes in BMD are defined as those exceeding the least significant change (LSC), which varies by measurement precision but is generally expected to be less than 5.0% at the total hip and 5.3% at the lumbar spine according to ISCD guidelines.²² Because measurement precision varies slightly between instruments, follow-up DXA scans are ideally conducted at the same facility with consistent equipment and protocols.

Risk Prediction Models

The World Health Organization (WHO) developed the Fracture Risk Assessment Tool (FRAX) in 2008 to predict the 10-year probability of a major osteoporosis-related fracture or hip fracture.²⁴ The computer-based FRAX algorithm is a probabilistic model that considers nine risk factors: (i) age, (ii) sex, (iii) BMI, (iv) fracture history, (v) parental fracture history, (vi) excessive alcohol usage, (vii) glucocorticoid use, (viii) smoking, and (ix) rheumatoid arthritis. These variables were found to be associated with osteoporosis, whether through estrogen loss, chronic inflammation, or disrupted bone turnover, in a meta-analysis of 12 population-based studies organized by the WHO.²⁴ The variables are entered in a binary (yes/no) format. Femoral

neck BMD represents an optional, but viable, FRAX input. Because BMD is not required for FRAX estimation, it can be used when DXA data are unavailable, including preliminary risk assessment of low-risk individuals. Competing risk of mortality is also incorporated, accounting for the chance that an elderly or severely ill individual might not live to sustain a fracture.

The next most commonly used and clinically accepted risk assessment tools, besides FRAX, are the Garvan fracture calculator and QFracture. Both are computerized prognostic algorithms. In comparison with FRAX, both these tools incorporate similar risk factors, with the exception of the competing risk of mortality. The Garvan fracture calculator considers five main risk factors, namely falls within the past 12 months, age, sex, fracture history, and BMD. QFracture places more emphasis on chronic diseases associated with secondary osteoporosis, such as diabetes mellitus, dementia, cardiovascular events, and Parkinson's disease; as such, it tends to estimate a higher 1-year fracture risk than the other clinically accepted risk prediction models.²⁵ The Garvan fracture calculator has also been found to slightly overestimate hip fracture risk over a 5-year period, but not other major osteoporotic fractures, compared to FRAX.²⁶

Simpler tools designed for the detection of low BMD, as discussed, include ORAI, OSIRIS, and OST. These clinically accepted predictive screening tools use fewer variables and are primarily intended for rapid clinical screening instead of detailed long-term fracture probability estimation.

LIMITATIONS TO CURRENT APPROACHES

Limitations of DXA

DXA is largely limited to bone density measurements. DXA essentially works by X-rays passing through the patient's tissue and bone; the attenuated X-rays, by applying the attenuation coefficient of bone, are used to determine bone mineral content (BMC). Using DXA methodology, BMD is calculated by dividing the BMC (in grams) by the total bone area (in cm²) covered. This measurement is technically referred to as areal BMD (aBMD), i.e., bone mass divided by unit of area. X-ray attenuation is influenced by the amount of tissue the beam passes through – a factor that cannot be fully captured in a 2D imaging system. Raw aBMD measurements are derived from X-ray intensity and may therefore vary with patient bone thickness (depth). For instance, larger bones are likely to have a larger raw aBMD measurement than smaller bones of identical bone density.²⁷ Volumetric BMD (vBMD) is the bone mass divided by the bone volume, and can be calculated using 3-dimensional scanners such as quantitative computed tomography (QCT). There is limited research that compares the effectiveness of aBMD with vBMD in predicting fracture risk. One study conducted by the Osteoporotic Fractures in Men (MrOS) Study Research Group found that while both vBMD and aBMD were associated with increased fracture risk, QCT had an advantage over DXA in predicting spinal fractures in men over the course of about ten years.²⁸ Similar results have also been observed in studies of postmenopausal women, potentially due to the greater sensitivity

of QCT, which improves the identification of particularly vulnerable individuals.²⁹ Another study conducted in Denmark indicated that a lower BMD in the hip and forearm was associated with fracture risk, but not the BMD of the lumbar spine, in both men and women.³⁰ These findings may indicate that DXA underestimates fracture risk, particularly for vertebral fractures, based on aBMD scores. Despite the advantage of 3D QCT methods in predicting spinal/vertebral fractures and even predicting cortical thickness, these methods also may not accurately capture cortical porosity³¹ and cortical thickness,³² both critical indicators of bone strength. Partial volume effects and limited resolution capabilities both contribute to QCT limitations.

There are other factors to keep in mind for BMD measurements; collagen and marrow do not come into consideration, only bone mineral, or hydroxyapatite (HA), the inorganic bone material. Degradation of organic bone material, especially type-1 collagen, has been directly linked to fracture risk.³³ Non-enzymatic glycation (NEG), or the stiffening of collagen associated with aging, diabetes, and other chronic diseases, also contributes to decreased bone quality. The contribution of factors other than BMD to fracture risk is evident in Figure 3, which shows BMD alongside the percentage of postmenopausal women who experience fractures over time. The figure demonstrates that fracture risk increases steadily with age by decade of life, even among individuals with similar BMD measurements. . Age-related increase in fracture risk is observed in both men and women, though risk rises more sharply in women after menopause. All else equal, between two individuals with identical BMD measurements, the older person has a greater chance of experiencing a fracture within ten years.^{34,35}

Figure 3 | Fracture probability in postmenopausal women by age and femoral neck BMD T-score.

Note: Omitted from this public version due to copyright. Please refer to Kanis et al.³⁵ for the original figure.

Finally, DXA neglects bone turnover markers, hormonal status, and genetic factors that influence the metabolic and integrative state of a person's bone. All these factors directly or indirectly affect bone quality, which is undetectable through DXA measurements alone. Bone remodeling and microarchitecture are also not entirely visible on the scans, leaving much to be desired. A 10-year follow-up study in 1311 postmenopausal women, completed in 2025, found that the relationship between DXA-assessed BMD T-score and fracture risk was not strictly linear.³⁶ The location of measurement, as found by this study, greatly influenced predictive ability; the best location was found to be the femoral neck. Even under optimal conditions, the overall predictive accuracy of DXA was moderate, with reported area under the curve (AUC) values around 0.7.³⁶ It is plausible, though not yet fully established, that these DXA limitations also apply to the male population.

Limitations of Prediction Models

Risk prediction models such as FRAX, ORAI, Garvan Fracture, and QFracture rely on a limited set of clinical risk factors as input variables. Input variables are the same for all patients, regardless of sex or ethnic background. These clinical risk factors result from large databases that served as a foundation for developing the statistical prediction models. FRAX was developed from epidemiological data collected over the course of several years. Other than the nine risk factors that constitute FRAX variables, other risk factors were considered and excluded by the WHO consensus due to inconsistent or insufficient data for global standardization, as the tool was developed for international application. These include biomarkers of bone turnover, stroke, diabetes, calcium nutrition, and falls.³⁷ However, during the studies conducted, diabetes was not strictly divided between Type 1 and Type 2; recent studies show that Type 2 diabetes is indeed a clinical risk factor for fracture risk and secondary osteoporosis.³⁸ Type 2 diabetes is associated with normal or elevated BMD, yet results in increased fracture risk, indicating a change in bone quality independent of bone mass. Calcium nutrition was excluded from the risk factors constituting FRAX assessment based on a meta-analysis that concluded that low milk intake was not significantly associated with hip fracture risk.³⁹ This finding suggests that self-reported calcium dietary intake may not serve as an adequate indicator of true blood calcium levels. Additional research confirms that calcium and vitamin D supplements may not have a direct impact on fracture risk.⁴⁰ However, this approach primarily reflects dietary intake rather than systemic homeostasis, which is influenced by endocrine regulation and other metabolic factors. For example, disturbances in calcium-regulating hormones such as calcitonin may affect bone remodeling and fracture susceptibility, especially in postmenopausal populations. Calcitonin has been shown to significantly reduce fracture risk in postmenopausal osteoporotic women.⁴¹

FRAX is largely built to predict hip fractures, and has been found to underestimate the risk of other major osteoporotic fractures (MOFs) – i.e., vertebral, humeral, and wrist fractures – in postmenopausal women.²⁶ In addition, the clinical use of FRAX as a predictive tool has not been shown to reduce fracture incidence in the long term, including both primary and secondary outcomes, particularly for non-hip fractures.⁴² Garvan Fracture and QFracture have not demonstrated significantly better performance. Both Garvan Fracture and FRAX were found to have moderate ability to predict major osteoporotic fractures other than hip fractures, while Garvan consistently overestimates hip fracture prevalence.²⁶ In turn, QFracture has been found to underestimate the risk for all fractures. QFracture is less widely calibrated than FRAX, as it is exclusively calibrated to the UK population, where it was developed and externally validated in a cohort similar to that used for its internal validation.⁴³ As a whole, these three models assume largely additive effects, which limits their ability to capture nonlinear interactions and population-specific variation in fracture risk.

Risk Prediction Method	Input Variables	Output	Population	Limitations
FRAX	Age, sex, BMI, fracture history, parental fracture, smoking, alcohol, glucocorticoids, rheumatoid arthritis, optional femoral neck (hip) BMD	Fracture risk (10-year)	Adults (global)	Assumes additive risk; underestimates non-hip fractures; has a limited impact on reducing fracture incidence
OSTA	Age, body weight	Osteoporosis risk	Asian adults	Excludes other variables; specific to the Asian population
ORAI	Age, body weight, current estrogen use	Osteoporosis risk	Postmenopausal women	Excludes other variables; specific to postmenopausal women
Garvan	Age, BMD, fall history	Fracture risk	Adults 50+	Not suitable for premenopausal or younger male populations
QFracture	Age, medical conditions, BMI, medications	Fracture risk	Adults (UK population)	Trained on a very narrow population pool

Table 1 | Comparison of clinically approved risk assessment methods.

In addition to risk prediction models, bone biopsy can provide information about the patient's bone microarchitecture and clinical biomarkers of osteoporosis. Bone biopsy, the gold standard for microarchitectural assessment, can provide more structural detail than most 2D or 3D scans, but is often considered only as a last resort because of its invasive nature, limited availability, and associated complications. Biopsies are usually reserved for patients who have already experienced fractures with the possibility of underlying issues such as bone cancers or blood cancers.⁴⁴

POSSIBLE SOLUTIONS

Biological & Clinical Data Integration

Incorporating both biological and clinical risk factors into determining osteoporosis risk reflects a shift toward more integrated approaches to fracture prediction and early disease detection. Advances in high-dimensional profiling attempt to transition beyond single-factor models toward integrated risk prediction utilizing proteomics, genomics, metabolomics, and transcriptomics. Individually, single risk factors provide limited insight, but collectively they can enable a more universal representation of osteoporosis-related fracture risk. A study from 2020 integrated data from 61 Caucasian women with high BMD and 58 with low BMD.⁴⁵ Correlation analysis methods resulted in 199 identified BMD-associated genetic loci and five distinguished biomarkers with causal effects. Genetic variants, including genes from the TNF/TNFR signaling network, were found to affect biomarkers through a web of complex biological/pathophysiological mechanisms. As previously discussed, genes from the TNF/TNFR signaling network, especially the osteoprotegerin gene (TNFRSF11B), have been historically associated with osteoporosis risk. These findings are particularly compelling; however, they should be interpreted with caution given the small sample size and limited genetic diversity, and require further experimental validation.. Additional studies stress the role of immune-bone interactions, the bone-muscle axis, and metabolic reprogramming in osteoporosis, linking conditions such as muscle degeneration and osteosarcopenia to bone loss and fracture risk.⁴⁶ These interconnected multi-system processes may shape long-term osteoporosis susceptibility. Inflammatory and epigenetic factors are believed to contribute to early-life processes that predispose individuals to osteoporosis later in life. Such studies transition the view of osteoporosis as a strictly bone disease to a “multisystem disorder involving bone–muscle and cross-disease interactions”.⁴⁶

Genome-wide association studies (GWAS) have contributed to the identification of about 1100 loci influencing BMD.⁴⁷ The integration of genetic and transcriptomic risk factors with clinical variables (including sex, age, BMI, and fracture history) might improve the performance of fracture risk prediction models. Reduced expression of the osteoblast differentiation-regulating transcription factors RUNX2, SOST, and Osterix has been reported in elderly male patients with osteoporosis.⁴⁸ Combined with oxidative stress and inflammation, disruption of bone resorption pathways has been linked to bone loss.⁴⁹ Several studies have developed predictive models using established clinical and epidemiological risk factors derived from population-based datasets. A 2024 risk prediction study found that combining genetic and clinical factors, rather than relying exclusively on clinical factors, modestly increased explained variance in bone mineral density by 2.3% for the femoral neck and 3.0% for the lumbar spine.⁵⁰ Researchers unified mixed-sex models with sex as a predictor variable, rather than constructing separate male and female prediction models. The authors concluded that incorporating specific genetic variants yields more targeted results than approaches based solely on GWAS. The reported

study was conducted on a British white population, and the prediction models performed poorly with other ethnic groups.⁵⁰ Similarly, genetic risk factors correlated with osteoporosis in populations of European descent failed to predict femoral neck fractures in Chinese populations.⁵¹ As such, genetic information may need to be trained on diverse populations or stratified based on ancestry-specific groups.

Figure 4 | Integrative omics analysis of osteoporosis biomarkers.

Note: Omitted from this public version due to copyright. Please refer to Zhao et al.⁴⁷ for the original figure.

Figure 4 highlights the contribution of multiple omics layers to osteoporosis-related biomarkers, demonstrating how genomic, transcriptomic, proteomic, and metabolomic factors shape disease risk. The figure emphasizes the interconnected nature of these systems, highlighting the complexity of osteoporosis as a multifactorial disorder.

Machine Learning Models

Machine learning (ML) models can identify complex, nonlinear relationships with high-dimensional data that would be difficult or impossible for traditional analytical capabilities to capture. ML has emerged in recent years as a strategy to detect osteoporosis risk through the integration of biological and clinical factors. Recent study review reports have documented the experimental use of artificial intelligence (AI) in the analysis of bone properties, osteoporosis classification, fracture detection, and risk prediction.⁵² However, not all ML classifiers are created equal. Model learning approaches can be classified into supervised and unsupervised models. Supervised models, which include classifiers such as logistic regression, neural networks, and random forest, among others, are trained on labeled input-output data to predict outcomes and can be used for risk prediction. Unsupervised models are not trained on labeled data and are instead programmed to find patterns within unlabeled data.

FEATURE	SUPERVISED LEARNING	UNSUPERVISED LEARNING
Data Type	Labeled data	Unlabeled data
Purpose	Predict outcomes	Identify patterns
Complexity	Less complex, structured	More complex, exploratory
Training Data Requirement	Requires large labeled datasets	Works well with raw data

Table 2 | Comparison of supervised and unsupervised machine learning.

Note: Adapted from ref.⁵³

Supervised models have shown promising performance in multiple studies to date. To date, the most common models used in studies to predict osteoporosis risk include support vector machines (SVMs), k-nearest neighbour (KNN), logistic regression, tree-based models (including decision trees and random forests), and artificial neural networks (ANN), including deep neural networks (DNN). Predictive algorithms can be constructed using these models individually or in hybrid combinations. Researchers from Guangzhou, China, constructed predictive algorithms using eight supervised models to predict osteoporosis risk from patient blood indicators.⁵⁴ The patient samples were collected via venous blood draw from 1560 male and female patients who had undergone DXA assessment between 2022 and 2023. The logistic regression model especially yielded strong performance, with area under the curve (AUC) values up to 0.835 and sensitivity of 80.9%; in contrast, the Osteoporosis Self Assessment Tool for Asians (OSTA), a commonly used screening tool for Asians that calculates osteoporosis risk based on age and body weight, has previously reported an AUC of 0.736 and a sensitivity of 73.1%.⁵⁴ While not directly comparable due to different input variables and differing model complexity, findings suggest superior performance of the machine learning model. The researchers used the SHapley Additive exPlanations (SHAP) method, a feature attribution method, to interpret results, and found uric acid, alkaline phosphatase, hemoglobin counts, and neutrophil counts as significant features in addition to clinical risk factors. Researchers suggest the future use of logistic regression model calculators as accessible pre-screening tools prior to DXA evaluation. Tree-based models, including decision trees and random forests, have demonstrated variable performance across studies. Decision trees are often ranked lower than more advanced machine learning models for predicting BMD,^{55,56} likely due to their relative simplicity and tendency to overfit. Random forest models have been shown to outperform decision trees,

support vector machines (SVMs), and other baseline classifiers in terms of sensitivity, AUC, and other performance criteria.⁵⁶

In addition to the more traditional supervised models, deep learning (DL), based on artificial neural networks, is one of the most powerful tools for risk detection, especially when combined with imaging data. Deep learning transitions further beyond risk factor integration to the identification of previously unrecognized relationships in data. Researchers found that deep learning improves detection accuracy when dealing with unstructured data compared to simpler ML algorithms.⁵⁷ DL models can be classified by neural network architecture and include convolutional neural networks (CNNs), multilayer perceptrons (MLPs), generative adversarial networks (GANs), autoencoders, and transfer learning (TL). CNN and TL models have been found to generally perform best for classification and detection of bone structures, including bone mineral density and microarchitecture,⁵⁸ while MLPs are used for tabular data-based prediction. An MLP deep learning model constructed and trained on clinical features yielded an AUC of 0.851 in US populations and 0.827 AUC in Korean populations for the femoral neck BMD prediction.⁵⁵ Results were even stronger for the prediction of total femur BMD. When the local interpretable model-agnostic explanations (LIME) algorithm, another feature attribution method, was applied to interpret the DL model, clinical features not typically emphasized in osteoporosis risk assessment – such as obesity, lower socioeconomic status, and high uric acid and alkaline phosphatase levels – were found to be associated with low BMD.

Figure 5 | Performance of selected supervised machine learning models in predicting osteoporosis in men and women, using body weight and age as input factors.

Note: Omitted from this public version due to copyright. Please refer to Qiu et. al.⁵⁹ for the original figure.

Unsupervised models are less geared toward risk prediction, yet still play a role in osteoporosis risk studies, especially when it comes to feature selection. Unsupervised learning can support feature selection by identifying underlying structure in high-dimensional data. Cluster analyses have been used to distinguish molecular subtypes of osteoporosis based on ferroptosis-related gene expression in secondary diabetic osteoporosis.⁶⁰ These subgroup distinctions can subsequently guide feature selection and supervised modeling.

FUTURE IMPROVEMENTS

Limitations of Current Machine Learning Approaches

Machine learning approaches come with a unique set of limitations and are relatively new in clinical development. No approaches thus far have been approved for clinical use. In addition, many studies intended to substantiate ML efficacy lack external validation.⁵² Recent attempts to

externally validate a random forest model for osteoporosis prediction using US and Chinese datasets were successful, but the small external cohort size limits generalizability.⁶¹ In addition, omics-derived disease predictors lack external validation and generalizability. Small cohort size and homogenous populations are also characteristic features of many multi-omics studies, and can lead to discrepancies between studies of omics approaches.⁴⁶ Most studies involving osteoporosis biomarkers utilize cross-sectional datasets or single time-point data, thereby lacking valuable insight into temporal metabolic dynamics that shape disease development. Because of the dynamic nature of biomarkers, cross-sectional studies can only give limited insight into causal effects.

Overfitting – i.e., overanalyzing training data to the point where noise becomes a problem – represents a fundamental issue, especially in more basic models. Overfitting causes low generalizability, especially when relying on small or non-representative datasets. When compared to nine other ML models, random forest performed best on the training set and thereby exhibited the most problematic overfitting phenomenon.⁶² High training AUC values driven from localized patterns can weaken real-world performance, where isolated events and random noise are inherently present. In turn, the near-perfect training AUCs can result in less-than-ideal testing results.⁶³ Incomplete or unstandardized datasets add to this problem and create an even more skewed gap between training datasets and independent, external datasets.

Inconsistencies in data collection and measurements of clinical factors can create subtle, but significant, discrepancies. As discussed earlier, BMD measurements can vary slightly from instrument to instrument; definitions of BMI may also vary between data collection protocols.⁶⁴ Most studies have been done on European populations, and with genetic factors playing a significant role in osteoporosis prediction, these findings may not fully apply to individuals from other racial and ethnic backgrounds.⁶⁴ GWAS also includes most data from largely European-ancestry populations. Data gaps, incomplete datasets (including those missing bone turnover markers or key biomarkers), small sample sizes in genomic studies, and the complexity of combining genomic with phenotypic data all contribute to the lack of generalizability. Finally, class imbalance in clinical datasets may bias models toward majority classes, thereby reducing sensitivity to high-risk patients.

DL models present notable challenges, particularly regarding interpretability. The “black box” problem refers to a lack of transparency in how algorithms assess relationships between variables. In other words, it can be difficult to understand exactly how an artificial neural network generates predictions and makes decisions with respect to data. Insufficient transparency can reduce clinical trust and present a significant barrier for the integration of DL models into healthcare systems.⁶⁵

Advanced Predictive Methodology & Optimization

Steps are being taken to bridge limitations and bring ML models closer to clinical use. Multimodal integration involves the combination of heterogeneous data types in order to

improve diagnostic precision. Researchers intend to develop predictive models that include genetic features, blood biomarkers, imaging data, and clinical risk factors.⁵⁷ Multimodal integration can yield the benefits of imaging-based assessments, such as DXA, with biological and clinical indicators, creating a more comprehensive representation of disease for clinical interpretation. When image-based ML classifiers incorporated 3D high-resolution CT scans (HR-pQCT scans) with DXA-measured BMD and clinical risk factors, fracture risk prediction achieved an AUC of 0.90.⁶⁶ In comparison, under the same conditions, the classifier relying only on DXA-measured BMD and clinical risk factors alone produced an AUC of around 0.71. The combined 3-dimensional imaging and AI can provide insight into bone microarchitecture that clinical methods to date have failed to fully capture.

Longitudinal studies are necessary for tracking biomarkers over time. Some longitudinal studies have been performed to track osteoporosis and the bone aging process, but most have an inherent cross-sectional design – i.e., they do not follow the same individuals continuously.⁴⁶ To help fill in this gap, researchers such as Yuan et al. have designed long-term studies to identify clinically relevant molecular signatures associated with osteoporosis.⁶⁷ The three-timepoint multimodal study conducted in China on 366 male and female individuals with 2- and 4-year follow-ups resulted in the identification of two pertinent molecular phenotypes, each associated with relevant bone turnover markers. The age range of the individuals was not specified; however, the cohort consisted of postmenopausal women and middle-aged men. In the two phenotypes, one was characterized by low hip BMD and the other by low lumbar spine BMD. The two subtypes were also found to respond differently to interventions such as calcium supplementation, suggesting that phenotypes in the future may dictate treatments for low BMD. These findings emphasize the wealth of insightful data that can be gained from true follow-up longitudinal studies.

Hybrid interpretable models and deep learning approaches attempt to balance transparency with advanced predictive performance — i.e., combine the interpretability of supervised learning approaches with the analytical power of deep learning methods. Hybrid models attempt to address the limitations caused by the “black box” issue. As early as 2022, researchers attempted to combine XGBoost, a decision tree model, with a deep neural network to predict fracture risk.⁶⁸ More recent approaches involve the integration of advanced computer vision models designed to detect spatial patterns. When computer vision models are combined with neural network architectures, they can improve the sensitivity and specificity of osteoporosis detection compared to traditional computer vision approaches. For example, a hybrid model utilizing combined convolutional and transformer-based architectures outperformed nine older computer vision models in specificity, precision, and accuracy for bone osteoporosis detection.⁶⁹ In addition to hybrid models, feature attribution methods, including SHAP and LIME, as discussed earlier, can assign values to input variables in deep learning models and thereby trace the impact of each variable. This can increase transparency and help address the “black box” problem. Research into feature attribution methods is still ongoing.

Altogether, this combination of technologies is promising for enabling earlier recognition of osteopenia and osteoporosis in a clinical setting using imaging and patient data alone.

Machine learning methods have shown promising advances toward early prediction and detection of osteoporosis. However, for clinical integration to be achieved, further work is necessary. Reliable methods must be tested through long-term, externally validated studies that include diverse populations to ensure generalizability. Transparency, accuracy, and standardization of both data collection and model evaluation are essential beyond initial model development. These elements are what can facilitate the transition of predictive models into integrated clinical use.

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FIGURES

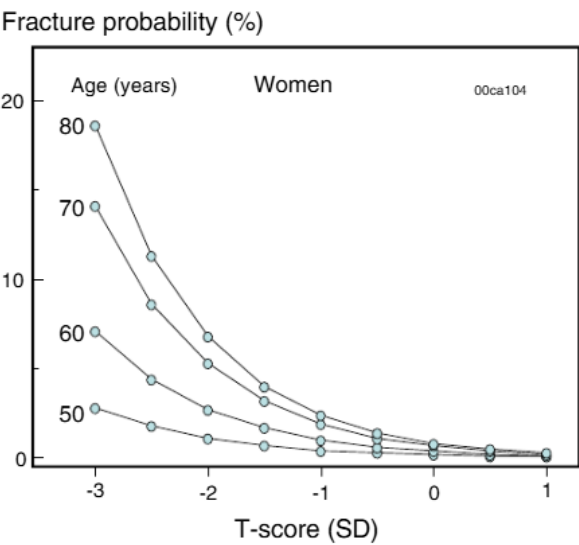


Figure 1
Fracture probability by age and femoral neck BMD T-score

Note: Adapted from “European guidance for the diagnosis and management of osteoporosis in postmenopausal women,” by J. A. Kanis et al., 2012, Osteoporosis International.³⁵

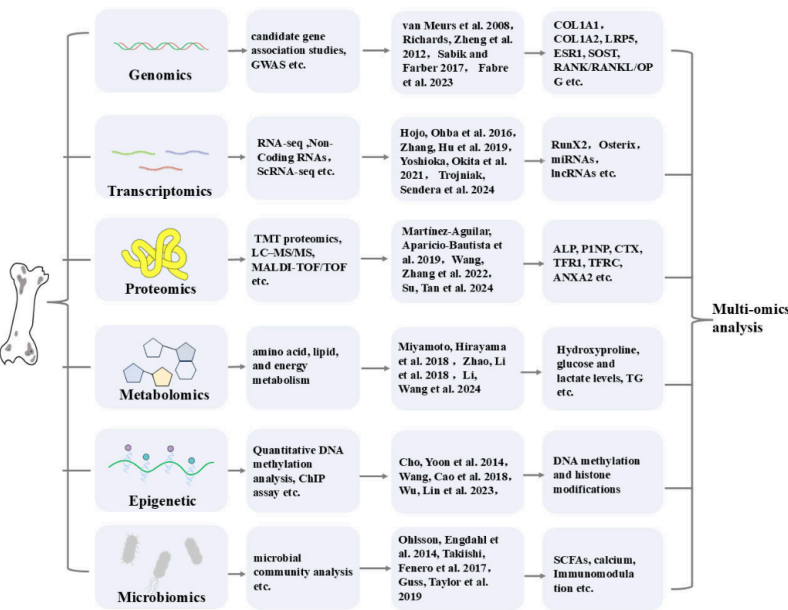


Figure 2

Integrative omics analysis of osteoporosis biomarkers

Note: Adapted from “Beyond Bone Loss: A Biology Perspective on Osteoporosis Pathogenesis, Multi-Omics Approaches, and Interconnected Mechanisms,” by Y. Zhao et al., 2025, Biomedicines.⁴⁷